

1. Let V be a real inner product space. A set $S = \{\mathbf{w}_1, \dots, \mathbf{w}_k\} \subseteq V$ is called *orthogonal* if it doesn't contain the zero vector, and if

$$\langle \mathbf{w}_i, \mathbf{w}_j \rangle = 0 \text{ for all } i \neq j.$$

- (a) Prove that every orthogonal set is an independent set.
- (b) Let $S = \{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ be an orthogonal set and suppose that

$$\mathbf{v} = a_1 \mathbf{w}_1 + a_2 \mathbf{w}_2 + \dots + a_k \mathbf{w}_k.$$

In this case, show that $a_i = \langle \mathbf{v}, \mathbf{w}_i \rangle / \langle \mathbf{w}_i, \mathbf{w}_i \rangle$ for all i .

- (c) If $\mathbf{w}$ is a nonzero vector and $\mathbf{u} = \mathbf{w} / \|\mathbf{w}\|$, show that $\|\mathbf{u}\| = 1$. We say that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an *orthonormal set* if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

If $\mathbf{u}_1, \dots, \mathbf{u}_k$ is orthonormal and $\mathbf{v} = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k$, show that $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$ for all i .

2. Let W be a vector space and let $U, V \subseteq W$ be subspaces.

- (a) Prove that $U \cap V$ is a subspace, but $U \cup V$ need not be a subspace.
- (b) Define the sum of subspaces by

$$U + V = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in U, \mathbf{v} \in V\}.$$

Prove that $U + V$ is a subspace of W .

- (c) Suppose that $U \cap V = \{\mathbf{0}\}$. In this case, show that for all vectors $\mathbf{w} \in U + V$ there exists a **unique** expression $\mathbf{w} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in U$ and $\mathbf{v} \in V$. [Hint: Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then ...]
- (d) If $W = U + V$ and $U \cap V = \{\mathbf{0}\}$ then we say that W is the *direct sum* of U and V , and we write

$$W = U \oplus V.$$

In this case, show that $\dim W = \dim U + \dim V$. [Hint: Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ be a basis of U and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis of V . Prove that $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis of W .]

3. Let V be a real inner product space and let $U \subseteq V$ be a subspace. We define the *orthogonal complement* of U as follows:

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

- (a) Prove that $U^\perp$ is a subspace of V .
- (b) Prove that $U \cap U^\perp = \{\mathbf{0}\}$.
- (c) If V is finite dimensional, prove that $V = U + U^\perp$. Combining this with (b) shows that $V = U \oplus U^\perp$, and hence

$$\dim V = \dim U + \dim U^\perp.$$

[Hint: Suppose that $\dim U = k$ and $\dim V = n$. From the Gram-Schmidt process (see below), it is possible to find an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_k$ for U such that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is a basis for U . Show that $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ are in $U^\perp$ and use this to show that every $\mathbf{v} \in V$ can be expressed as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$.]

4. Let V be a finite-dimensional real inner product space and let $U \subseteq V$ be a subspace. From Problems 2 and 3 we know that for every vector $\mathbf{x} \in V$ there exists a unique pair of vectors $\mathbf{x}_U \in U$ and $\mathbf{x}_{U^\perp} \in U^\perp$ such that

$$\mathbf{x} = \mathbf{x}_U + \mathbf{x}_{U^\perp}.$$

Define two functions $P_U, P_{U^\perp} : V \rightarrow V$, called the *orthogonal projections* onto U and $U^\perp$, by setting $P_U(\mathbf{v}) := \mathbf{v}_U$ and $P_{U^\perp}(\mathbf{v}) := \mathbf{v}_{U^\perp}$. Suppose that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an orthonormal basis for U , meaning an orthogonal basis with $\|\mathbf{u}_i\| = 1$ for all i . In this case, prove that

$$P_U(\mathbf{x}) = \langle \mathbf{x}, \mathbf{u}_1 \rangle \mathbf{u}_1 + \langle \mathbf{x}, \mathbf{u}_2 \rangle \mathbf{u}_2 + \dots + \langle \mathbf{x}, \mathbf{u}_k \rangle \mathbf{u}_k$$

and

$$P_{U^\perp}(\mathbf{x}) = \mathbf{x} - P_U(\mathbf{x}) = \mathbf{x} - \sum_{i=1}^k \langle \mathbf{x}, \mathbf{u}_i \rangle \mathbf{u}_i.$$

[Hint: See Problem 1(b).]

Challenge Problem (Gram-Schmidt Procedure). Let V be a finite-dimensional inner product space over $\mathbb{R}$. The *Gram-Schmidt procedure* is used to convert any basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ into an orthogonal basis $\mathbf{w}_1, \dots, \mathbf{w}_n$. We inductively define

$$\begin{aligned} \mathbf{w}_1 &:= \mathbf{v}_1, \\ \mathbf{w}_2 &:= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1, \\ \mathbf{w}_3 &:= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 - \frac{\langle \mathbf{v}_3, \mathbf{w}_2 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2, \\ &\text{etc.} \end{aligned}$$

If we wish to have an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ then we can take $\mathbf{u}_i := \mathbf{w}_i / \|\mathbf{w}_i\|$.

(a) Prove by induction that, for all $k \geq 1$,

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_k\} \text{ and } \mathbf{w}_k \neq \mathbf{0}.$$

[Remark: We need $\mathbf{w}_k \neq \mathbf{0}$ so we can divide by $\langle \mathbf{w}_k, \mathbf{w}_k \rangle$ in the definition of $\mathbf{w}_{k+1}$.]

(b) Prove that $\langle \mathbf{w}_i, \mathbf{w}_j \rangle = 0$ for all $i \neq j$.

(c) If we define the subspace $W_k := \text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$, prove that

$$\mathbf{w}_{k+1} = P_{W_k^\perp}(\mathbf{v}_{k+1}) \quad \text{for all } k \geq 1.$$

[Hint: Use the formula from Problem 4 and the fact that $W_k = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$.]

(d) The Gram-Schmidt procedure can also be applied in infinite-dimensional vector spaces. Consider the function space $L^2[0, 1]$ with inner product

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x) dx.$$

If we apply Gram-Schmidt to the infinite sequence $1, x, x^2, \dots$, then we obtain an infinite sequence of polynomials $p_0(x), p_1(x), p_2(x), \dots$ called the *shifted Legendre polynomials*. Compute the first three Legendre polynomials and check that they are orthogonal with respect to the inner product.